M731–Partial Differential Equations HOMEWORK – Fall 2026

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All homework should be uploaded in Canvas by the due date and time.

SET 1: Due Thursday 09/24/2026 by 7pm

From McOwen's Book: read the introduction & Sections 1.1d) and 1.3

From McOwen's Book Section 1.1 do: 1, 2, 3, 4a)b), 6a) 9.

SET 2. Due: Thursday 10/8/2026 by 7pm

Additional Problems. Read carefully the first Handout posted on the course webpage about inhomogeneous transport equations. You should be familiar with this from undergraduate PDE at least when $d = 1$. Here this is in any dimension $d \geq 1$. Note that in the Handout I wrote n instead of d for the dimension so adjust the notation to the additional problems 1 and 2 below.

(1) Write down an explicit formula for a function u solving the inhomogeneous initial value problem

$$\begin{cases} u_t + b \cdot \nabla u = f & \text{on} \quad \mathbb{R}^d \times (0, \infty) \\ u = g & \text{on} \quad \mathbb{R}^d \times \{t = 0\} \end{cases}$$

where $f = f(x,t)$, $f : \mathbb{R}^d \times [0, \infty) \to \mathbb{R}$, $c \in \mathbb{R}$, $b \in \mathbb{R}^d$ and $g : \mathbb{R}^d \to \mathbb{R}$ are all given. (Hint. Solve first the homogeneous problem (i.e. $f = 0$) using the method of characteristics (as in class or McOwen section 1.1a)). Then use the Fundamental Theorem of Calculus to write $u(x,t) - g(x - bt)$ for (x,t) along a characteristic ($g(x - bt)$ is the homogeneous solution) as an appropriate integral of f. At the end u will be the sum of the homogeneous solution plus an integral term.)

(2) Write down an explicit formula for a function u solving the initial value problem

$$\begin{cases} u_t + b \cdot \nabla u + c\, u = 0 & \text{on} \quad \mathbb{R}^d \times (0, \infty) \\ u = g & \text{on} \quad \mathbb{R}^d \times \{t = 0\} \end{cases}$$

where $c \in \mathbb{R}$, $b \in \mathbb{R}^d$ and $g : \mathbb{R}^d \to \mathbb{R}$ are given.

This problem below is a basic problem from undergraduate Analysis. Might be useful to think about it for balls in $\mathbb{R}^d$ first to gain intuition.

(3) Let f be a continuous function on an open set $D \subset \mathbb{R}^d$ such that

$$\int_{D_0} f(x)\, dx = 0 \qquad \text{for all} \quad D_0 \subset D.$$

Prove that then $f \equiv 0$ on D.

From McOwen's Book Section 2.1: 1, 2, 7

Note: I. will do related problem to 7 in the first half of class on Tuesday 10/06 to show an an example failure if stability -continuous dependance of the solution on its data)

From McOwen's Book: Read carefully **McOwen Section 2.2a) pages 49-51** classifying equations of second order in $\mathbb{R}^2$. You have learned this classification (hyperbolic, parabolic, elliptic) in undergraduate PDE but here the conditions are expressed in terms of characteristic curves. After reading carefully, rewrite this section yourselves filling any gaps and explaining the steps and conclusions. Write or type clearly your notes and hand them to me on Thursday.

SET 3. Due: Thursday 10/15/2026 by 8pm

From McOwen's Book Section 2.2: Read carefully Section 2.2b). Then **do** 1.

Additional Problems.

1) a) Find the characteristics of the PDE

$$y^2 u_{xx} - 2y u_{xy} + u_{yy} = u_x + 6y,$$

and determine if elliptic, parabolic or hyperbolic.

b) Then find the canonical form and use it to find the solution u first in the ξ and η variables and then in the x and y variables.

2) a) Find the characteristics of the PDE

$$x u_{xx} + (x - y) u_{xy} - y u_{yy} = 0, \qquad x > 0, y > 0,$$

and determine if elliptic, parabolic or hyperbolic.

b) Then show that it can be transformed into the canonical form

$$(\xi^2 + 4\eta) u_{\xi\eta} + \xi u_\eta = 0$$

for ξ and η are suitably chosen canonical coordinates. and use this to obtain the general solution in the ξ and η variables.

SET 4. Due: 10/22/2026 by 9pm

From McOwen's Book Section 3.1: 1, 4, 5

From McOwen's Book section 3.3: Read and rewrite carefully the statement and the proof of Theorem 2 which proves that the wave equation has finite propagation speed! To prove (50) from (48) use the Hint in the back of the book. Write in the details.

SET 5. Due: 10/29/2026 by 9pm

From Robert McOwen's Book Section 3.2: 1, 2, 2*, 3, 5, 6a)

Hint for Problem 2: First show that if $y = (y_1, y_2, y_3)$ is a point in the unit sphere $\mathbb{S}^2$ then

$$\int_{\mathbb{S}^2} y_j \, d\sigma(y) \, = \, 0 \qquad\qquad j = 1, 2, 3$$

where as always $d\sigma$ is the area surface element. This can be proved by an explicit calculation using spherical coordinates or also by symmetry, splitting for each j, the integral over the sphere $\mathbb{S}^2$ into the two integrals for the half spheres $y_j \geq 0$ and $y_j \leq 0$ and showing the two integrals cancel out).

Problem 2* Same as problem 2 but with initial conditions $u(x, 0) = 0$ and $u_t(x, 0) = x_2$.

Hint for Problem 5 Follow the hint in the back of the book and define $u(x_1, x_2, x_3) = \cos(\frac{m}{c} x_3) v(x_1, x_2, t)$. Then prove by a direct calculation u satisfies the $3d$ (linear homogeneous) wave equation. Hence u can be represented in terms of g and h using Kirchhoff's formula. Assume first for simplicity that $g = 0$ and write the explicit formula for u in this case. Then set $x_3 = 0$ and proceed as in the 'method of descent' (parametrize the two halves of $\mathbb{S}^2$ corresponding by graphs $y_3 = \pm\sqrt{1 - (y_1^2 + y_2^2)}$) to obtain a formula for v which should look like:

$$v(x_1, x_2, t) = C\, t \int_D \frac{\cos\left(mt\sqrt{1 - (y_1^2 + y_2^2)} \right) h(x_1 + cty_1, x_2 + cty_2)}{\sqrt{1 - (y_1^2 + y_2^2)}} \, dy_1 dy_2$$

where $D := \{(y_1, y_2) : \, y_1^2 + y_2^2 \leq 1 \}$.

Finally drop the assumption that $g = 0$ to get the general formula.

Hint for Problem 6a) The decay in t in $3d$ is due to waves spreading out in space on expanding spheres $\partial B(x, ct)$ as $t \to \infty$. This bound reflects the *dispersion* of waves. Assume first $g = 0$ and use Kirchhoff's formula in conjunction with the fact that h is bounded (i.e. $|h(x)| \leq M$ for some $M > 0$) and compactly supported (hence $h(x) = 0$ if $|x| \geq R$ for some large $R > 0$). From these (plus change of variables) one can prove that

$$|u(x, t)| \leq \frac{M}{Ct} \, \text{area}(\partial B(x, ct) \cap B(0, R))$$

where $C > 0$ depends on the area of the unit sphere and the speed c. Next show that the area of the *spherical cap* $\partial B(x, ct) \cap B(0, R)$ can be bounded by some absolute constant times R^2- and hence by a quantity that is independent of x and t. Next note that if $h = 0$ instead then both terms in g in Kirchhoff's formula can be treated similarly as the previous case (do it!). Finally, for $g, h \in C_0^\infty(\mathbb{R}^3)$ we have the sum of 3 terms all of which we know how to treat.

Note related to part 6b). In $2d$ there is 'one less direction' than in $3d$ for waves to spread out so intuitively we expect the amplitude of the waves to decay slower as time increases. And indeed, in $2d$ there is decay bound of the form $|u(x, t)| \leq \frac{C}{\sqrt{t}}$ but this is harder to prove.

In $1d$, however, as we can clearly see from D'Alembert's formula there is no decay at all as $t \to \infty$.

Form Robert McOwen's Book Section 3.3: 1, 2, 4

Do but do not turn in Write down with full details Problem 5 in Section 3.3 .

SET 6. Due: 11/05/2026 by 9pm

Form McOwen's Book Section 2.3: 4, 8, 10, 11c).

Additional Problems on Distributions.

(1) Prove the Remark at the bottom of page 10 of the Notes on Distributions. This is a generalization of Pb 8 in section 2.3.

A generalized version of this statement is the following problem. Explain how would you do it:

(1') Let $\{f_j\} \in L^1(\mathbb{R}^n)$ be a sequence of nonnegative functions such that

$$\int_{\mathbb{R}^n} f_j(x)\,dx \to 1 \quad \text{as} \quad j \to \infty \quad \text{and for any } a > 0,$$
$$\int_{|x|>a} f_j(x)\,dx \to 0 \quad \text{as} \quad j \to \infty.$$

Let F_{f_j} be the distribution defined by f_j. Prove that $F_{f_j} \to \delta$ in $\mathcal{D}'$.

(2) For δ the Dirac delta distribution, find the k-th (weak) derivative $\delta^{(k)}$ for any $k \geq 1$.

(3) Find the derivative in the sense of distributions of the $\operatorname{sgn} x$ (the function on $\mathbb{R}$ defined to be equal to 1 for $x > 0$, equal to 0 for $x = 0$ and equal to -1 for $x < 0$).

(4) Compute $\frac{d}{dx}(\log|x|)$ on $\mathbb{R}$ in the sense of distributions.
Recall that the principal value of $\frac{1}{x}$ (pv $\frac{1}{x}$) is defined as

$$\langle \mathrm{pv}\frac{1}{x}, \phi\rangle = \mathrm{pv}\int_{\mathbb{R}} \frac{\phi(x)}{x}\,dx := \lim_{\varepsilon\to 0^+}\int_{|x|>\varepsilon} \frac{\phi(x)}{x}\,dx$$

(5) Prove that the Dirac delta distribution δ_{x_0} with point mass at $x_0 \in \mathbb{R}^n$ (fixed) is not given by a locally integrable function. In other words prove that there does not exist any $f \in L^1_{loc}(\mathbb{R}^n)$ such that

$$\langle \delta_{x_0}, v\rangle = \langle f, v\rangle \qquad \text{for all} \quad v \in C_0^\infty(\mathbb{R}^n)$$